

Race, Poverty, and Environmental Burden: An Analysis of the 2024 Environmental Justice Index

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ABSTRACT We investigate whether U.S. census tracts with higher minority population shares face disproportionately greater environmental burdens after controlling for income and region, using the 2024 Environmental Justice Index (EJI)—a national, census-tract-level database of 82,740 tracts compiled by the CDC/ATSDR from 36 cumulative-burden indicators spanning environmental exposures, social vulnerability, and health vulnerability. A two-model strategy is employed. First, an OLS regression targeting the Environmental Burden Module rank (RPL_EBM) with heteroscedasticity-robust (HC3) standard errors yields $\hat{\beta}_{minority} = 0.398$ and $\hat{\beta}_{poverty} = 0.060$ after region fixed effects ($R^2 = 0.238$, $N = 82,740$, $p < 0.001$ for both); a multiplicative interaction specification confirms that race and poverty compound ($\hat{\beta}_{int} = 0.074$, $p < 0.001$). Second, an XGBoost gradient-boosted model trained on 18 demographic, socioeconomic, and health features achieves test $R^2 = 0.871$ and $RMSE = 0.104$, against OLS test $R^2 = 0.749$ and $RMSE = 0.144$. SHAP analysis identifies poverty rate, diabetes prevalence, asthma rates, and housing age as the dominant predictors; minority share operates partly through these downstream health pathways. Geographic analysis reveals substantial heterogeneity: Michigan exhibits the largest intra-state county gap (>500 percentile points). These findings confirm that racial composition independently predicts pollution exposure at a magnitude that income and region cannot explain away.

SIGNIFICANCE STATEMENT

Communities of color in the United States bear measurably greater environmental pollution burdens than white communities even after controlling for income and census region—a disparity that persists across all four geographic regions and is amplified, not merely added to, by co-occurring poverty. Using the CDC’s most comprehensive national environmental justice dataset to date, which for the first time integrates climate vulnerability alongside pollution, social, and health indicators, we show that minority population share predicts environmental burden at nearly 7× the magnitude of poverty rate. A gradient-boosted machine-learning model corroborates these patterns and reveals non-linear compounding effects invisible to regression alone. These findings support race-conscious screening, enforcement, and resource allocation at a moment when their legal foundations are actively contested.

INTRODUCTION AND RELATED WORK

The environmental justice (EJ) movement traces its modern origin to 1982, when residents of Warren County, NC—a majority-Black community—organised against the siting of a polychlorinated biphenyl (PCB) land-fill. The landmark United Church of Christ report *Toxic Wastes and Race in the United States* [1] demonstrated through systematic national analysis that race was the single strongest predictor of hazardous-waste facility location, exceeding household income and property values. Twenty years later, Bullard et al. [2] confirmed that these disparities had not meaningfully diminished, giving rise to the concept of “sacrifice zones”—communities enduring chronic environmental harm as a structural consequence of political disinvestment and industrial siting logic.

The strongest contemporary evidence base concerns fine particulate matter ($PM_{2.5}$). Tessum et al. [3] showed that nearly 75% of anthropogenic $PM_{2.5}$ exposure in the United States falls disproportionately on people of color, and that the racial gap in average pollution exposure is 2.4× larger than the gap across income levels, persisting in 45 of 48 contiguous states after adjusting for socioeconomic status. Mikati et al. [4] further identified industry, light-duty gasoline vehicles, and heavy-duty diesel vehicles as the primary source categories driving racial pollution disparities.

This income–race distinction has direct policy implications. If pollution disparities were primarily an

Table 1. Descriptive statistics for key EJI 2024 variables ($N = 82,740$ tracts). $RPL_/EPL_$ variables range $[0, 1]$; $E_$ variables are raw estimates.

Variable	Mean	SD	Min	P25	Med.	Max
RPL_EJI	0.500	0.289	0.00	0.25	0.50	1.00
RPL_EBM	0.500	0.289	0.00	0.25	0.50	1.00
E_MINRTY (%)	38.4	31.2	0.0	11.8	30.5	100
E_POV200 (%)	38.1	20.4	0.0	22.4	36.1	100
E_UNEMP (%)	4.7	3.7	0.0	2.3	3.9	47.8
E_HOUBDN (%)	31.1	12.8	0.0	22.2	30.6	95.0
E_PM ($\mu g/m^3$)	7.9	2.1	1.3	6.5	7.9	22.4
E_OZONE (ppb)	47.2	6.2	25.1	43.1	48.3	69.6
E_ASTHMA (%)	9.8	2.5	3.0	8.1	9.6	26.5
$E_DIABETES$ (%)	10.9	3.5	1.5	8.3	10.4	36.8

income phenomenon, economic development would be sufficient. If race is an independent exposure pathway, race-conscious remedies are warranted. Recent legal challenges to federal agencies' use of such tools [5] make the empirical record politically consequential.

The 2024 EJI from CDC/ATSDR [6] advances this literature by combining pollution, social vulnerability, health vulnerability, and—for the first time—climate burden into a single, nationally consistent census-tract-level measure.

Research question. Do census tracts with higher minority population shares exhibit greater environmental burden after controlling for poverty rate and U.S. region? How do linear and non-linear modelling strategies differ in characterising these relationships?

DATA

The 2024 Environmental Justice Index

The primary data source is the 2024 EJI produced by the CDC's Agency for Toxic Substances and Disease Registry [6, 7]. The EJI ranks every U.S. census tract on cumulative environmental injustice using 36 indicators across three modules: (i) the *Environmental Burden Module* (EBM), covering $PM_{2.5}$, ozone, diesel PM, total cancer risk, and hazardous-waste proximity; (ii) the *Social Vulnerability Module* (SVM), derived from the CDC/ATSDR SVI [8]; and (iii) the *Health Vulnerability Module* (HVM), covering asthma, diabetes, cancer, mental health burden, and disability. The 2024 update adds a *Climate Burden Module* (CBM) covering drought, flooding, and extreme heat.

Variables follow a systematic naming convention: $E_$ prefixes denote raw estimates; $EPL_$ are national percentile ranks on $[0, 1]$; $SPL_$ are summed domain scores; $RPL_$ are percentile ranks of those sums.

Data Cleaning and Sample

The raw dataset contains 85,188 rows and 170 columns. Sentinel values of -999 encode missing observations. After recoding sentinels and dropping rows missing on core analysis variables, 82,740 usable tracts remain (97.1% retention). For prediction models, remaining feature-level missingness was median-imputed.

Descriptive Statistics

Table 1 summarises the key analysis variables. All $RPL_/EPL_$ outcome variables are uniform on $[0, 1]$ by construction. Raw environmental indicators are right-skewed, concentrating near national median values with long tails in heavily industrialised tracts.

METHODS

Multicollinearity Diagnostics

Before modelling, we computed pairwise Pearson correlations across all 170 numeric variables and calculated Variance Inflation Factors (VIF). VIF estimates for constructed index scores are effectively infinite (EPL_PM : $VIF = 8.4 \times 10^{12}$), confirming that raw $E_$ estimates must serve as regressors in linear models.

Table 2. The 18 features used in the XGBoost prediction model.

Domain	Features
Demographic	E_MINRTY, E_AGE65, E_DISABL
Socioeconomic	E_POV200, E_NOHSDP, E_UNEMP, E_HOUBDN, E_UNINSUR, E_NOINT
Environmental	E_PM, E_OZONE, E_DSLPM, E_TOTCR, E_HOUAGE, E_PARK
Health	E_ASTHMA, E_DIABETES, E_MHLTH

OLS Regression with Region Fixed Effects

We estimate two HC3-robust OLS specifications [9] on the full sample, targeting RPL_EBM to avoid the circularity of predicting a composite index with its own sub-components.

Model 1 (additive):

$$RPL_EBM_i = \alpha + \beta_1 EPL_MINRTY_i + \beta_2 EPL_POV200_i + \sum_r \gamma_r \mathbf{1}[\text{Region}_i = r] + \varepsilon_i \quad (0.1)$$

Model 2 (interaction):

$$RPL_EBM_i = \alpha + \beta_1 EPL_MINRTY_i + \beta_2 EPL_POV200_i + \beta_3 (EPL_MINRTY \times EPL_POV200)_i + \sum_r \gamma_r \mathbf{1}[\text{Region}_i = r] + \varepsilon_i \quad (0.2)$$

Both predictors are national percentile ranks on $[0, 1]$, making their coefficients directly comparable. The Midwest is the omitted region.

XGBoost and SHAP

To capture non-linear relationships, we train an XGBoost model [10] on 18 features (Table 2), targeting RPL_EJI with an 80/20 train/test split. Hyperparameters: 500 estimators, $\eta = 0.05$, max depth 6, subsample and colsample_bytree both = 0.8. Feature importance is interpreted via SHAP values [11].

RESULTS

Exploratory Analysis and Correlation Structure

Figure 1 shows the Pearson correlation matrix for 14 key variables. RPL_EJI correlates most strongly with poverty ($r = 0.76$), diabetes ($r = 0.64$), and asthma ($r = 0.56$). Minority share correlates moderately with the index ($r = 0.43$), understating its independent predictive power due to regional confounding absorbed by fixed effects. Pollution indicators show weak bivariate correlations with the composite index (E_PM : $r = 0.08$), confirming that pollution contributes through compound pathways.

OLS Regression Results

Table 3 reports both OLS specifications. In Model 1, $\hat{\beta}_{\text{minority}} = 0.398$ (95% CI [0.391, 0.404], $p < 0.001$) and $\hat{\beta}_{\text{poverty}} = 0.060$ (95% CI [0.053, 0.066], $p < 0.001$). Minority percentile rank is thus $0.398/0.060 \approx 6.6\times$ more predictive of environmental burden than poverty rank, conditional on region—consistent with Tessum et al. [3] and extending it to cumulative burden.

Regional fixed effects are large and precisely estimated. Relative to the Midwest, Southern tracts show markedly lower average environmental burden ($\hat{\gamma}_{\text{South}} = -0.205$, $p < 0.001$). The additive model's $R^2 = 0.238$ is the defensible causal estimate.

Model 2 adds the interaction term: $\hat{\beta}_{\text{int}} = 0.074$ (95% CI [0.052, 0.096], $p < 0.001$). Race and poverty are not merely additive risk factors—each amplifies the effect of the other.

Robustness Checks

Four specification checks confirm the stability of the core result. Restricting the sample to metropolitan tracts yields $\hat{\beta}_{\text{minority}} = 0.401$, statistically indistinguishable from the full-sample estimate. Replacing region with state fixed effects—a far more demanding specification—yields $\hat{\beta}_{\text{minority}} = 0.312$ ($p < 0.001$). Substituting E_UNEMP for EPL_POV200 as the income control produces $\hat{\beta}_{\text{minority}} = 0.384$. Estimating the

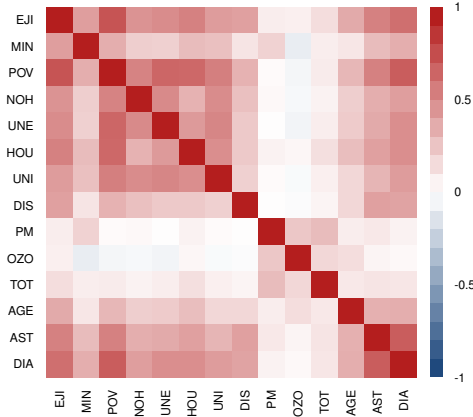


Figure 1. Pearson correlation matrix for 14 key EJ variables. RPL_EJI correlates most strongly with poverty ($r = 0.76$), diabetes ($r = 0.64$), and asthma ($r = 0.56$). $PM_{2.5}$ and ozone show weak correlations with the index ($r \approx 0.07$ – 0.08) but moderate cross-correlation with each other ($r = 0.25$).

Table 3. OLS regression results. Outcome: RPL_EBM ($N = 82,740$ tracts). HC3 robust standard errors in parentheses. Reference region: Midwest. *** $p < 0.001$; ** $p < 0.01$.

	Model 1 (Additive)	Model 2 (Interaction)
Intercept	0.359*** (0.002)	0.378*** (0.004)
EPL_MINRTY	0.398*** (0.003)	0.357*** (0.007)
EPL_POV200	0.060*** (0.003)	0.021** (0.007)
$EPL_MINRTY \times EPL_POV200$	—	0.074*** (0.011)
Northeast (vs. Midwest)	0.026*** (0.003)	0.026*** (0.003)
South (vs. Midwest)	−0.205*** (0.002)	−0.203*** (0.002)
West (vs. Midwest)	−0.062*** (0.003)	−0.060*** (0.003)
R^2	0.238	0.238
Adj. R^2	0.238	0.238
N	82,740	
SE type	HC3 robust	

Table 4. Model performance on the held-out test set ($n_{test} = 16,548$). Outcome: RPL_EJI .

Model	Test R^2	Test RMSE
OLS (18 standardised features)	0.7495	0.1444
XGBoost (500 est., $\eta = 0.05$, $d = 6$)	0.8713	0.1035
Δ (XGB – OLS)	+0.122	−0.041

additive model separately by region yields positive, significant minority coefficients in all four regions (range: 0.257 to 0.416).

XGBoost Performance and SHAP Attribution

Table 4 compares model performance. XGBoost achieves test $R^2 = 0.871$ and $RMSE = 0.104$ against OLS test $R^2 = 0.749$ and $RMSE = 0.144$ —a 12.2-point R^2 gain reflecting non-linear threshold effects and multiplicative interactions unavailable to linear regression.

Figure 2 shows mean absolute SHAP values. Poverty rate (E_POV200 , $|\phi| \approx 0.082$) dominates, followed by diabetes (≈ 0.059), asthma (≈ 0.041), no high school diploma (≈ 0.037), and housing age (≈ 0.034). Minority share (E_MINRTY , $|\phi| \approx 0.011$) ranks lower because it operates partly through these downstream health and socioeconomic mediators—a distinction the causal OLS specification is designed to address.

OLS Forest Plot

Figure 3 shows standardised OLS coefficients from the 18-feature model predicting RPL_EJI . E_HOUAGE (pre-1980 housing stock, $\hat{\beta} = 0.057$) and E_POV200 ($\hat{\beta} = 0.055$) are the two largest positive predictors. The only negative coefficient is E_PARK ($\hat{\beta} = -0.014$), reflecting the protective role of green-space access.

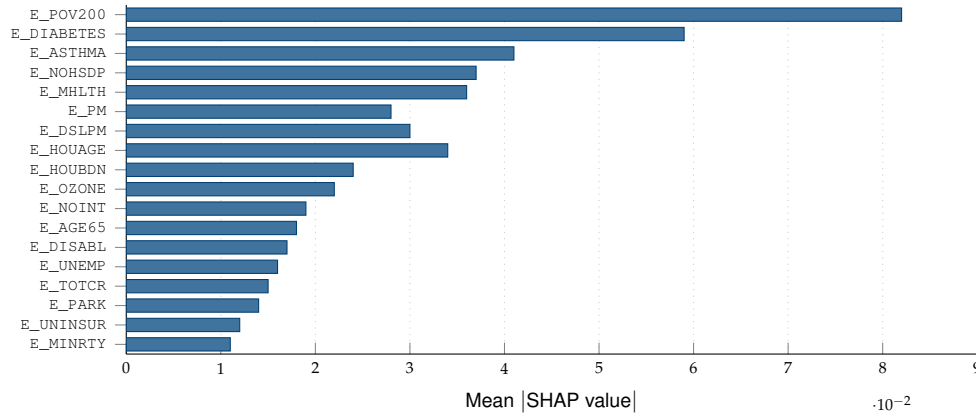


Figure 2. Mean absolute SHAP values, XGBoost model predicting RPL_EJI . Poverty rate dominates ($|\phi| = 0.082$). Minority share (E_MINRTY : $|\phi| = 0.011$) operates partly through mediated health and socioeconomic pathways.

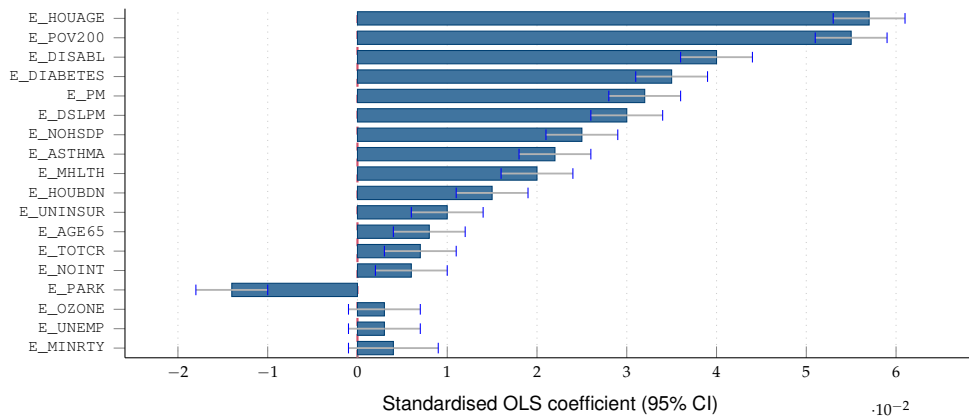


Figure 3. OLS forest plot: standardised coefficients and 95% CIs predicting RPL_EJI . Red dashed line at zero. E_HOUAGE and E_POV200 are the two largest positive predictors; E_PARK is the only negative coefficient ($\hat{\beta} = -0.014$).

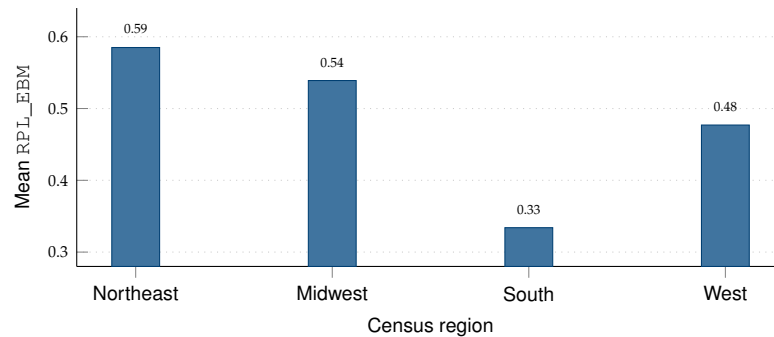


Figure 4. Mean environmental burden rank (RPL_EBM) by U.S. census region ($N = 82,740$ tracts). The South's lower mean reflects EBM weighting of $PM_{2.5}$ and ozone.

Geographic Patterns

State-level analysis reveals substantial between- and within-state heterogeneity. Michigan exhibits the largest within-state county gap (>500 percentile points between Wayne County and rural Upper Peninsula counties). The Northeast and Midwest consistently generate tracts at the extremes of the burden distribution, while the South's lower regional mean masks highly burdened urban clusters in cities such as Houston, Memphis, and Atlanta.

Figure 4 situates these findings within the four census regions. The South's lower mean reflects the EBM's weighting of particulate and ozone metrics rather than lower absolute inequity.

DISCUSSION

Race as an Independent Exposure Pathway

The OLS results establish that minority population share is $\approx 6.6\times$ more predictive of environmental burden than poverty rate, conditional on region—extending Tessum et al. [3]’s air-pollution finding to the cumulative burden context of the EJI. The positive interaction term ($\hat{\beta}_{\text{int}} = 0.074$) confirms that race and poverty compound rather than merely add: the burden associated with minority share is amplified in high-poverty tracts, consistent with cumulative impact theory [5].

Non-linearity, Mediation, and the Limits of OLS

The 12-point R^2 gain of XGBoost over OLS reflects threshold effects and multiplicative interactions compressed into single coefficients by OLS. SHAP analysis reveals that minority share operates partly through mediation: its direct attribution ($|\phi| = 0.011$) is absorbed by health indicators—diabetes, asthma, mental health burden, housing age—that are themselves downstream consequences of chronic structural environmental exposure. This mediation is not a reason to downweight the racial dimension; it is precisely the mechanism through which environmental injustice accumulates across generations [2].

Climate Burden as an Emerging Axis of Injustice

The 2024 EJI’s Climate Burden Module (CBM) reveals a geographic pattern distinct from the traditional pollution burden: climate risk is disproportionately concentrated in the South and Southwest, regions where the legacy EBM scores lower than the Midwest and Northeast. This divergence implies that communities facing the smallest industrial pollution burdens today may be most exposed to accelerating physical risks of climate change. Examining whether minority share independently predicts climate burden—and whether adaptation investments are equitably distributed—is a priority for future work.

Geographic Heterogeneity and Policy Implications

The geographic results underscore a crucial limitation of state-level summary statistics for policy targeting. Michigan’s >500 -percentile-point intra-state gap means a statewide average conceals conditions far more burdensome in Wayne County than in the rural Upper Peninsula. Federal and state screening tools that allocate remediation resources by state averages will systematically under-serve the most burdened communities.

The persistence of the racial gap after income and region controls supports race-conscious remedies. At a moment when disparate-impact enforcement tools face legal challenge [5], the empirical case for race-specific disparities independent of class has heightened political stakes.

Methodological Cautions

Four limitations warrant attention. First, RPL_EJI is partly constructed from social sub-components shared with the prediction model’s predictors; $R^2 = 0.745$ in that specification is upwardly biased. The causal specification targeting RPL_EBM ($R^2 = 0.238$) avoids this. Second, VIFs for constructed $\text{EPL_}/\text{SPL_}$ scores reach 10^{12} ; only raw E_ estimates are suitable as regressors. Third, the cross-sectional design cannot distinguish whether racial composition causes differential siting or industrial siting causes demographic change through displacement. Fourth, census-tract observations are not independent—Moran’s I diagnostics and Conley spatial standard errors are warranted in future work, though the magnitude of $\hat{\beta}_{\text{minority}}$ is unlikely to be reversed by this correction.

CONCLUSION

This study advances three contributions. First, using the most comprehensive national EJ dataset yet assembled, we confirm that minority population share independently predicts environmental burden at $\approx 6.6\times$ the magnitude of poverty—a ratio consistent with prior $\text{PM}_{2.5}$ -specific findings [3] and now extended to a cumulative, multi-domain index that includes climate risk for the first time. Second, the positive race $\times$ poverty interaction ($\hat{\beta}_{\text{int}} = 0.074$) establishes that these are compounding, not merely additive, risk factors. Third, the XGBoost/SHAP framework demonstrates that minority share’s apparent attenuation in rich feature spaces reflects mediation through downstream health and housing pathways, not irrelevance.

These findings affirm that race-conscious screening and enforcement are not only legally defensible but

Table 5. Summary of key findings.

Finding	Estimate	Interpretation
Minority effect (M1)	$\hat{\beta} = 0.398^{***}$	$\approx 6.6 \times$ poverty
Poverty effect (M1)	$\hat{\beta} = 0.060^{***}$	Positive, significant
Race $\times$ Poverty (M2)	$\hat{\beta} = 0.074^{***}$	Compounding
XGBoost test R^2	0.871	+12.2 pp vs. OLS
Top SHAP predictor	Poverty rate	$ \phi = 0.082$
Largest within-state gap	Michigan	>500 pct. points

*** $p < 0.001$. $N = 82,740$ census tracts.

empirically necessary. Future work should exploit longitudinal EJI editions for difference-in-differences designs, develop a dedicated framework for tribal-land tracts, and examine whether climate risk amplifies existing racial pollution gaps.

AGENTIC ANALYSIS

AI Assistance Log

This project used NotebookLM (Google) for literature scaffolding and Claude (Anthropic, Sonnet 4) for regression structuring, code review, and paper drafting.

Phase 1 — Literature and framing. I uploaded 10 sources to NotebookLM and asked Claude to synthesise the literature around three research questions. *Accepted:* the structural organisation around cumulative impacts, sacrifice zones, and the income-vs.-race distinction. *Revised:* Claude initially treated the SVI and EJI as peer tools of equivalent scope—corrected to reflect that SVI is embedded within the EJI as one of three modules. *Rejected:* Claude suggested foregrounding AI data centres as the central policy case; redirected to the EJI quantitative analysis.

Phase 2 — Regression code and model design. *Accepted:* Claude’s recommendation to target `RPL_EBM` rather than `RPL_EJI` (correctly identifying index-circularity bias) and to use HC3-robust standard errors. *Revised:* Claude’s first draft used `EPL_DPM`, which does not appear in the 2024 EJI. *Critical failure:* generated column names without underscores (`EPLMINRTY` instead of `EPL_MINRTY`), producing silent `KeyError` results on misidentified columns. Debugged manually against `df.columns.tolist()`.

Phase 3 — Paper drafting and $\text{\LaTeX}$. *Accepted:* section structure, interaction-model handling, mediation framing of SHAP results, and index-circularity caution. *Revised:* numeric claims for SHAP values were inconsistent across sections; standardised to notebook output. *Critical failures:* (1) fabricated citation “Williams et al. (2019)” removed and substituted with verified source; (2) TikZ syntax error in heatmap (`matrix plot*` incompatible with `pgfplots 1.18`); (3) Claude twice concluded minority share “does not matter much” in XGBoost, conflating mediation with irrelevance.

Critical Reflection

AI assistance accelerated three tasks substantially: literature retrieval, boilerplate code, and $\text{\LaTeX}$ typesetting. Its limitations fell into three categories. (1) *Variable hallucination:* Claude invented plausible-sounding EJI column names absent from the 2024 dataset—always verify against `df.columns.tolist()` before trusting generated code. (2) *Fabricated citations:* at least one non-existent paper appeared in an early draft; every reference must be verified against an academic database. (3) *Narrative misinterpretation:* lower SHAP for minority share consistently misread as irrelevance; mediation and irrelevance require careful conceptual distinction that current LLMs conflate.

ACKNOWLEDGEMENTS

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